2d. Content for the Programming Project

The Programming Project provides an opportunity for learners to demonstrate their practical ability in the skills outlined in the specification, supporting the learning of Components 01 and 02. We have provided a cross mapping grid in Section 2f to highlight where Component 01 and 02 may be delivered holistically as part of the Programming Project.

There must be clear evidence submitted to show that each learner has received 20 hours of timetabled delivery for the Programming Project.

The Programming Project requires learners to use skills from Component 01 and Component 02 to create a solution to a set problem. They will code their solution in a suitable programming language. The solution must be tested to ensure they solve the stated problem. Learners must create a suitable test plan with appropriate test data.

The code must be suitably annotated to describe the process. Test results should be annotated to show how these relate to the code, the test plan and the original problem.

Learners will need to provide an evaluation of their solution based on the test evidence.

Learners should be encouraged to be innovative and creative in how they solve the task.

Learners must use a suitable high-level text-based programming language such as:

- Python
- C family of languages (for example C#, C++, etc.)
- Java
- JavaScript
- Visual Basic/.Net
- PHP
- Delphi
- BASIC.

Computational thinking is in essence the ability to model problems in a manner that makes them amenable to computational solutions; it is not simply instructions and actions. Computational thinkers are able to see algorithms, processes and data and know how to then implement them in their chosen language.

When developing a solution to the programming task, we would suggest using an iterative process, such as below:

- **Success criteria** – what key things must the solution contain?
- **Planning and design** – the solution is broken down and suitable designs created
- **Development** – the iterative development with code explanations
- **Testing and remedial actions** – a log of successful tests including correcting any errors
- **Evaluation** – a review of the success criteria that have been met.

This process will allow learners to demonstrate the key elements of computational thinking:

- **Thinking abstractly** – removing unnecessary detail from the problem, and Control and Data abstraction
- **Thinking ahead** – identifying preconditions and inputs and outputs
- **Thinking procedurally** – identifying components of problems and solutions
- **Thinking logically** – predicting and analysing problems
- **Thinking concurrently** – spotting and using similarities.